

Learning Program: Intelligence Tradecraft

Course Title	Intelligence Tradecraft
Purpose	Intelligence Tradecraft equips students with the conceptual frameworks and practical tools required to think and operate like intelligence analysts. The purpose is to bridge the gap between subject-matter expertise and problem-solving tradecraft. Security professionals often hold deep knowledge of threats, technologies, or environments, but may not have been trained in the structured analytic methods that convert knowledge into foresight and actionable judgment. This course closes that gap by building a cognitive foundation in reasoning, biases, and analytic frameworks, then applying these concepts through hands-on exercises in hypothesis testing, structured techniques, and uncertainty communication. Students learn to work with intelligence from collection through to finished products: extracting information, building structured knowledge, applying analytic rigor, and communicating findings with appropriate confidence. The course embeds habits of critical thinking, intellectual curiosity, and self-awareness that enable analysts to navigate complexity and uncertainty. Critically, it teaches the analytic tradecraft and structured thinking that underpins effective data-driven analysis, ensuring that technical skills serve analytical judgment rather than replace it.
Learning Outcomes	By the end of this course a student should be able to: 1. Explain the Data-Information-Knowledge-Wisdom framework and apply it to intelligence problems. 2. Understand how to use who what where when how why & whither in analysis. 3. Be aware of biases and heuristics and know how to manage them. 4. Understand the fundamentals of ontology. 5. Build a basic intelligence ontology and use it to capture target knowledge. 6. Understand the different types of reasoning used in analysis. 7. Understand and use words of estimative probability. 8. Know how to assess intelligence collection for source credibility and reliability. 9. Write basic information reports. 10. Communicate uncertainty to decision makers 11. Apply basic structured analytic techniques.

	12. Perform hypothesis-led analysis.		
Target Audience	- Intelligence professionals. - Investigators and analysts.		
Course Size	12 Students	Duration	2 Days
Delivery Method	Remote / Hybrid / In Person	Prerequisites	Fundamentals of Intelligence

Session No.	Learning Outcomes	Learning Content (Topics)	Delivery Methods	Resources
1 (70 mins) <i>Data- Information- Knowledge- Wisdom</i>	Explain the Data-Information-Knowledge-Wisdom framework and apply it to intelligence problems.	- Traditional DIKW - DIWK adapted for Intelligence - Exercise: Where do you go at night? - Why this matters for analysts - Exercise: Practice identification	PowerPoint (live presentation) Moodle (exercises)	

Session No.	Learning Outcomes	Learning Content (Topics)	Delivery Methods	Resources
2 (60 mins) <i>Who What</i> <i>Where When</i> <i>How Why (and</i> <i>Whither)</i>	Understand how to use who what where when how why & whither in analysis.	- WWWWHW - Who - Where - When - Exercise: Find who what where - What - How - Why - Whither - Exercise: Analyse the report	PowerPoint (live presentation) Moodle (exercises)	
3 (30 mins) <i>Biases</i>	Be aware of biases and heuristics and know how to manage them.	- What are biases? - Confirmation Bias - Anchoring Bias - Availability Bias - Mirror Imaging	PowerPoint (live presentation) Moodle (exercises)	https://thedecisionlab.com/biases-index
4 (30 mins) <i>Heuristics</i>	Be aware of biases and heuristics and know how to manage them.	- Exercise: categorise this - Key Heuristics - Linking this to intelligence	PowerPoint (live presentation) Moodle (exercises)	https://thedecisionlab.com/biases-index

Session No.	Learning Outcomes	Learning Content (Topics)	Delivery Methods	Resources
5 (60 mins) <i>Introduction to Ontology</i>	- Understand the fundamentals of ontology. - Build a basic intelligence ontology and use it to capture target knowledge.	- What is ontology - Exercise: Everyday mapping - How do we use ontology in intelligence? - Exercise: Build a criminal intelligence ontology	PowerPoint (live presentation) Moodle (exercises)	
6 (60 mins) <i>Information Quality and Analytical Confidence</i>	- Understand and use words of estimative probability. - Know how to assess intelligence collection for source credibility and reliability.	- Assessing collected information - Exercise: Applying admiralty code - Words of estimative probability - Exercise: Assign words of estimative probability	PowerPoint (live presentation) Moodle (exercises)	

Session No.	Learning Outcomes	Learning Content (Topics)	Delivery Methods	Resources
7 (60 mins) <i>Induction and Deduction</i>	Understand the different types of reasoning used in analysis.	- Logic and reasoning - Inductive reasoning - Exercise: Inductive reasoning - Deductive reasoning - Exercise - Bringing it all together	PowerPoint (live presentation) Moodle (exercises)	
9 (30 mins) <i>Hypothesis-led analysis</i>	Perform hypothesis-led analysis.	- What are hypotheses? - Generating hypotheses - Testing hypotheses - Exercise: Build information needs	PowerPoint (live presentation) Moodle (exercises)	

Session No.	Learning Outcomes	Learning Content (Topics)	Delivery Methods	Resources
11 (60 mins) <i>Structured Analytic Techniques</i>	Apply basic structured analytic techniques.	- What are structured analytic techniques? - Key assumption check - Analysis of competing hypotheses - Exercise: Analysis of Competing Hypotheses - What If? Analysis	PowerPoint (live presentation) Moodle (exercises)	
10 (30 mins) <i>Information Reports</i>	- Write basic information reports. - Understand how to use who what where when how why & whither in analysis.	- What are information reports? - How to write information reports - Exercise: Write information reports	PowerPoint (live presentation) Moodle (exercises)	
12 (50 mins) <i>Communicating Uncertainty</i>	- Communicate uncertainty to decision makers - Understand and use words of estimative probability.	- From information quality to analytic confidence - Exercise: Assess Analytic Confidence - Communicating uncertainty to stakeholders - Exercise: Writing uncertainty	PowerPoint (live presentation) Moodle (exercises)	Canned intelligence examples for exercises, uploaded to Moodle.

Session No.	Learning Outcomes	Learning Content (Topics)	Delivery Methods	Resources
8 (120 mins) <i>Capstone Exercise</i>	- Explain the Data-Information-Knowledge-Wisdom framework and apply it to intelligence problems. - Understand the fundamentals of ontology. - Build a basic intelligence ontology and use it to capture target knowledge.	- Revision of previous topics - Introduction to the exercise - Exercise: Data to information - Exercise: Write information reports - Exercise: Building an ontology - Exercise: Information to knowledge - Exercise: Capturing target knowledge - Exercise: Write short report	PowerPoint (live presentation) Moodle (exercises)	Target cards and descriptions Scenario: Criminal intelligence

Session Plan 1: Data-Information-Knowledge-Wisdom

Course Title	Intelligence Tradecraft		
Session Title	Data-Information-Knowledge-Wisdom		
Purpose	This session establishes the foundational framework for distinguishing between observation and inference in intelligence analysis. Students will learn to identify data, information, knowledge, and wisdom as distinct epistemic layers, enabling them to build defensible assessments and communicate the certainty of their conclusions with precision.		
Learning Outcomes	Explain the Data-Information-Knowledge-Wisdom framework and apply it to intelligence problems.		
Resources & Equipment	PowerPoint (live presentation) Moodle (exercises)		
Preparation	Moodle exercise check; class admin.		
Participants	12	Session Duration	70 minutes

Time	Topic	Content and Activities	Resources
3 mins	Introduction	What we're going to be covering in this session.	
10 mins	Traditional DIKW	- Placeholder: during development, read original papers (Ackoff/Zeleny) and explain with details. - Mention that it isn't a single model, it's contested, but is useful for framing information management in a bunch of domains.	
15 mins	DIWK adapted for Intelligence	- The concepts behind how I frame the layers. - Talk about how they aren't clean breaks – there are edge cases for each transition, but the important point is making sure that you're recognising that there are changes as data moves to knowledge, and that there are	

Time	Topic	Content and Activities	Resources
		different epistemic conditions that need to be considered when presenting the results of your intelligence analysis. • Data as signal, raw stuff, through to things like images and logs. • Note that data doesn't refer to the stuff in databases, these can contain information and knowledge. Mention information bases and knowledge bases, but that they are a different framing. • Talk about information as being contextualised and observational, a fact made with characteristics like accuracy, confidence, and precision. • Knowledge as the product of inference about a subject. • Wisdom as an understanding of the world which facilitates the translation of information into knowledge (and data to information where the process isn't innate).	
20 mins	Exercise: Where do you go at night?	• This is an instructor-facilitated exercise. • The students are presented with a character, Chad Corporate, and information about his locations. • He is shown to be at home most nights between 0001-0500, so we can conclude that this is his home. • The nights he's not home are preceded by his visits to social events, restaurants and bars, suggesting an unsettled personal life. • Students take the information, use knowledge about the world, to conclude what Chad is up to. • We then pass forward in time, find him to be spending more time at a single other location. • We hypothesise that this is the location of a partner, Sophie Synergy. • We then find the information of who lives there. • She is also spending time at his home, reciprocal. • Fast forward and they're both living in the suburbs. • This exercise shows that by using data and information, combined with wisdom about the world, we can form conclusions about the subject of the analysis. • The wisdom layer needs to be explicit – students must present hypotheses about what data might be able to say. • Throw in a few red herrings.	
10 mins	Why this matters for analysts	• Not all intelligence is the same thing. • It's useful to understand how you step up all this stuff which is called intelligence.	

Time	Topic	Content and Activities	Resources
		• This framing gives signals for levels of certainty. • It forces epistemic accountability. • It allows you to structure your analysis better, helps to do the task so to speak.	
10 mins	Exercise: Practice identification	• Students are presented with a series of statements/scenarios • They must identify which DIKW layer each statement belongs to • Mix of difficulty levels – some obvious, some testing the Information/Knowledge boundary • Instructor reveals correct answers with brief reasoning • Reinforces the distinctions between observation and inference • Builds confidence in applying the framework before moving to next topic	
2 mins	Conclusion	• Tell em' what you told 'em.	

Session Plan 2: Who What Where When How Why (and Whither)

Course Title	Intelligence Tradecraft		
Session Title	Who What Where When How Why (and Whither)		
Purpose	This session teaches students to systematically decompose intelligence problems using the foundational questions of who, what, where, when, how, why, and whither. Students will learn to distinguish between entities, events, locations, and temporal markers as distinct analytical categories, enabling structured extraction of information from complex narratives.		
Learning Outcomes	Understand how to use who what where when how why & whither in analysis.		
Resources & Equipment	PowerPoint (live presentation) Moodle (exercises)		
Preparation	Moodle exercise check; class admin.		
Participants	12	Session Duration	60 minutes

Time	Topic	Content and Activities	Resources
3 mins	Introduction	- What we're covering. - Link it back to DIKW.	
5 mins	WWWVHW	- Introduce WWWVHW. - Aristotle, Sherlock Holmes. - Foundational framing for investigative work.	
5 mins	Who	- Who as in things in the world. - Distinguish between people and objects.	
5 mins	Where	Places as logical objects (Melbourne) – how we'll deal with them here.	

Time	Topic	Content and Activities	Resources
		Places as parameters (points, polygons) – lead into geospatial, mention that space is a continuum.	
5 mins	When	- Talk about sensible bins of time (days of week, dates, hours), logical objects. - Time as a continuum.	
5 mins	Exercise: Find who what where	- Students are given a SVO structured sentence, with potential time and space makers. - Extract the who and action, with the where and when. - For space and time, look at whether they're treating them as parameters or logical objects. - Examine the types of objects (the who) that can exist.	
5 mins	What	- What as events. - Can be composites of other events or observations. - Show as "nominalised event": as a noun phrase rather than SVO. - May run long, but how, why and whither will be covered lightly from here.	
5 mins	How	- How as method. - It is likely to be the product of analysis or hypothesis if in an intelligence product. - Hint that will be covered in more detail in hypothesis-led analysis session.	
5 mins	Why	- Why as motive - Talk about motive analysis. - Hint that will be covered in more detail in hypothesis-led analysis session	
5 mins	Whither	- Whither as an assessment of the future. - Talk about intelligence as making judgements about future state. - Included because of the expectation on intelligence analysts.	
10 mins	Exercise: Analyse the report	- Give student a comprehensive report. - Show events, methods, motive, assessment of the future. - Have students extract these from the narrative and present them as examples.	
2 mins	Conclusion	Tell 'em what you told 'em.	

Session Plan 3: Biases

Course Title	Intelligence Tradecraft		
Session Title	Biases		
Purpose	This session introduces cognitive biases that systematically distort intelligence analysis and provides strategies to recognize and mitigate their effects. Understanding these biases is essential preparation for the heuristics session that follows, and forms the foundation for hypothesis-led analysis and structured analytic techniques covered later in the course.		
Learning Outcomes	Be aware of biases and heuristics and know how to manage them.		
Resources & Equipment	PowerPoint (live presentation) Moodle (exercises) https://thedecisionlab.com/biases-index		
Preparation	Moodle exercise check; class admin.		
Participants	12	Session Duration	30 minutes

Time	Topic	Content and Activities	Resources
3 mins	Introduction	What we're going to cover.	
5 mins	What are biases?	- Your brain is amazing but it's lying to you. - We all have biases. - With structure and awareness we can manage them. - Example: gorilla suit basketball (or find another)	
5 mins	Confirmation Bias	It is a bias that has us focus on evidence that aligns to current beliefs.	

Time	Topic	Content and Activities	Resources
		• Critical in intelligence – we’re out looking for the things that have been set in requirements. The whole system has been set up to find these things. • Is an enormous challenge in environments that are focussed on intelligence challenges. • You can manage this by using alternative hypotheses and actively seeking disconfirming evidence.	
5 mins	Anchoring Bias	• Relying on the first piece of information we receive. • We interpret further information based on this. • Prevents us from critically considering evidence that diverts from this first anchoring. • Can manage by removal of initial information and seeing whether subsequent information is still consistent with the conclusion.	
5 mins	Availability Bias	• You overestimate the probability of memorable/recent events. • Isn’t necessarily related to the real probability of probability. • You can manage this by examining trends over time, grounding an assessment in evidence, supressing recency and testing whether the conclusion is still sound. • Link through to heuristic coming next session.	
5 mins	Mirror Imaging	• Projecting of own mindset onto an adversary. • “The mirror-imaging bias occurs when analysts project their own mindset onto others or assume that adversaries will act in the same manner as the United States.” Kwoun. • You can manage this through target and cultural context in intelligence analysis. • Thinking that the target is like you, quote McNamara, that they thought Vietnam was staunchly communist because the US framed it as an ideological struggle, when they were nationalist first and foremost.	
2 mins	Closing	• Heuristics after lunch. • This leads into hypothesis-led analysis. • Tell ‘em what you told ‘em.	

Session Plan 4: Heuristics

Course Title	Intelligence Tradecraft		
Session Title	Heuristics		
Purpose	This session explains how mental shortcuts (heuristics) enable rapid decision-making but can also produce systematic errors when applied inappropriately. Students will understand the relationship between heuristics and biases, recognise when heuristics are operating in their analysis, and learn how structured tradecraft can manage their impact.		
Learning Outcomes	Be aware of biases and heuristics and know how to manage them.		
Resources & Equipment	PowerPoint (live presentation) Moodle (exercises) https://thedecisionlab.com/biases-index		
Preparation	Moodle exercise check; class admin.		
Participants	12	Session Duration	30 minutes

Time	Topic	Content and Activities	Resources
3 mins	Introduction	- Refer back to biases (lunch). - Introduce heuristics, and how they're useful to humans.	
5 mins	Exercise: categorise this	Have psychologists find a good online demonstration of how people rapidly put things into categories.	Lean on IndependINT psychology knowledge.
15 mins	Key Heuristics	- Heuristic 1: Availability Heuristic (morning bridge) - Heuristic 2: Representativeness Heuristic – judging probability by similarity to stereotypes – use engineer/lawyer example from Olga.	Lean on IndependINT psychology knowledge.

Time	Topic	Content and Activities	Resources
		Heuristic 3: Anchoring and Adjustment Heuristic: start with an initial value, adjust insufficiently.	
5 mins	Linking this to intelligence	- You have heuristics for better or worse. - They can allow analysts to make judgements when, for example, trawling through vast rows of data, picking out that one lead instinctively. - However, they can also lead to faulty conclusions. - You need to be aware of heuristics so that you can know when they're occurring. - Once you know, you can apply tradecraft to manage their impact.	
2 mins	Closing	Tell 'em what you told 'em.	

Session Plan 5: Introduction to Ontology

Course Title	Intelligence Tradecraft		
Session Title	Introduction to Ontology		
Purpose	This session introduces ontology as the systematic framework for categorizing and structuring intelligence information. Students will learn how ontological design flows from analytical requirements and enables scalable data-driven analysis, including automated entity extraction and knowledge base construction.		
Learning Outcomes	Understand the fundamentals of ontology.		
Resources & Equipment	PowerPoint (live presentation) Moodle (exercises)		
Preparation	Moodle exercise check; class admin.		
Participants	12	Session Duration	60 mins

Time	Topic	Content and Activities	Resources
3 mins	Introduction	What we're going to cover.	
15 mins	What is ontology	- Ground in philosophical definition: Ontology is the study of existence and being, addressing questions about what is real and how existing things are categorised. - Be prepared to go deep on theory, to start heavy with the philosophical points. - Show a common ontology (Common Core Ontology). - Keep it light - Show how organisations extend ontologies into their domains.	

Time	Topic	Content and Activities	Resources
10 mins	Exercise: Everyday mapping	- Students are given a simple ontology. - Students are given a simple narrative, something like a classroom of kindergarten children. - They are to extract information from the narrative and place it into the ontology, mapping objects and relationships. - Should be light, bring it back from the heavy theory.	
20 mins	How do we use ontology in intelligence?	- Ontologies can be simpler than what we saw earlier. - You should build simple ontologies for your missions. - Requirements tell us our information needs. - If we need to know about a category of object, relationship, or attribute for collection, it needs to be accounted for in an ontology. - Once an ontology is established, we use it to create things like databases. - Show a flow: from requirements through collection to processing, how after collection is done the ontology takes back over in the processing phase. - Demonstrate how to handle complex objects, things like time and space. - Show entity extraction using ML or LLMs, how ontologies, when properly scoped and defined, can be used to automate analysis.	
10 mins	Exercise: Build a criminal intelligence ontology	- Students are given a narrative of a crime, perhaps a news article. - They have to extract the objects and relationships. - Use WWWWHW to extract the context. - Build out an ontology for the people involved: suspect, offender, witness, victim. - Create relationships between the objects and events. - Worked through as a group exercise. - This operationalises ontology.	
2 mins	Closing	Tell 'em what you told 'em.	

Session Plan 6: Information Quality and Analytical Confidence

Course Title	Intelligence Tradecraft		
Session Title	Information Quality and Analytical Confidence		
Purpose	This session introduces systematic frameworks for evaluating information quality and communicating analytical confidence in intelligence assessments. Students will learn to apply the Admiralty Code for rating sources and information and use Words of Estimative Probability to express likelihood judgments with precision and consistency.		
Learning Outcomes	Understand and use words of estimative probability. Know how to assess intelligence collection for source credibility and reliability.		
Resources & Equipment	PowerPoint (live presentation) Moodle (exercises)		
Preparation	Moodle exercise check; class admin.		
Participants	12	Session Duration	60 minutes

Time	Topic	Content and Activities	Resources
3 mins	Introduction		
20 mins	Assessing collected information	- Introduce the concept that the source reliability and the confidence in the information are two separate concepts. - Credibility is contextual: someone who is an expert in one thing may not be in another. - Credibility is built - Confidence in information is related both - Start with the admiralty code as standard approach.	

Time	Topic	Content and Activities	Resources
		- Run through the matrix. - Show students example collect (Tweets) of each. - Examine alternatives (for research, I want to be thorough). - Explain that these things can be best assessed statistically.	
10 mins	Exercise: Applying admiralty code	- Take tweets or news articles. - Assess source credibility and quality of information. - Assign admiralty codes. - In small groups first (4x3), then back as a group.	
15 mins	Words of estimative probability	- Go back to known knowns etc. - Reinforce that intelligence is a method of managing uncertainty. - History with Sherman Kent. - Show the words with percentages. - Explain that this is related to not just how you report things but how your audience receives it. - Look at some intelligence reports, how they use language. - Contrast this with other types of reporting, scientific with statistics of certainty, things like editorials with assertions only.	
10 mins	Exercise: Assign words of estimative probability	- Students are given a source collect, tweet or other. - They are given information reports about these tweets without words of estimative probability. - They are asked to choose the appropriate words to use. - Small groups first, then return to the main group.	
2 mins	Closing	- Tell 'em what you told 'em. - Day close to follow.	

Session Plan 7: Induction and Deduction

Course Title	Intelligence Tradecraft		
Session Title	Induction and Deduction		
Purpose	To establish foundational understanding of logical reasoning patterns that underpin all intelligence analysis, enabling analysts to think critically about how conclusions are formed and tested.		
Learning Outcomes	Understand the different types of reasoning used in analysis.		
Resources & Equipment	PowerPoint (live presentation) Moodle (exercises) *Find good online resources for students to use later.		
Preparation	Moodle exercise check; class admin. First lesson of the morning, make sure everyone is back.		
Participants	12	Session Duration	60 minutes

Time	Topic	Content and Activities	Resources
7 mins	Introduction	- Recap previous day. - Introduce the day. - Introduce the topic.	
5 mins	Logic and reasoning	- Why do philosophers spend so much time talking about how we reason? - The importance of understanding your reasoning patterns. - Different types of reasoning have different innate strengths and weaknesses. - We won't cover them all today, but we can make a start	

Time	Topic	Content and Activities	Resources
10 mins	Inductive reasoning	- Inductive reasoning, in its most basic form, starts with observations, identifies patterns, and then forms a generalisation. - Your scientists live here. - Examine these patterns in life: weather, science, business. - Demonstrate how induction is used in intelligence, with observations made through intelligence collection. - Focus on how the conclusions (generalisations) have to flow explicitly from the observations. Be aware of any hidden assumptions you are making. - Understand that correlation does not equal causation, and that you are likely forming a hypothesis that needs further testing.	*Find good online resources for students to use later.
10 mins	Exercise: Inductive reasoning	- Instructor led for first example. - Students are given a series of observations, simple ones, then asked to form conclusions. - Then break into small groups. - Use absurd examples, like opening umbrellas cause rain due to correlations. - Students produce the condition for testing a conclusion (look for times where it's raining but there are no umbrellas). - Back to the group, share conclusions.	
10 mins	Deductive reasoning	- Think your traditional philosophical logic. - Introduce prepositions. - Show basic A, B, therefore C. - Don't go too deep but show examples of sound and unsound syllogisms: must be both valid and true. - Show these with intelligence examples.	*Find good online resources for students to use later.
10 mins	Exercise	- Students break into groups. - Are given syllogisms. - Have to conclude whether they are sound or unsound.	
5 mins	Bringing it all together	- Through your analysis you'll move between different patterns of analysis. - Show the generalisation caused by induction feeding in as a preposition for deductive reasoning, using intelligence examples. - Induction and deduction are strong fundamental patterns of reasoning that you are using.	

Time	Topic	Content and Activities	Resources
		• It's important to recognise what patterns you're using so that you can be critical of your own thinking. • Once you start to do this, it will become natural.	
3 mins	Closing	• Tell 'em what you told 'em.	

Session Plan 8: Hypothesis-led Analysis

Course Title	Intelligence Tradecraft		
Session Title	Hypothesis-led Analysis		
Purpose	This module introduces hypothesis-led analysis as the organising framework for intelligence work, teaching students how to generate testable hypotheses from requirements and collection, and translate them into information needs that drive further analysis.		
Learning Outcomes	Perform hypothesis-led analysis.		
Resources & Equipment	PowerPoint (live presentation) Moodle (exercises) Intelligence requirement document		
Preparation	Moodle exercise check; class admin.		
Participants	12	Session Duration	30 minutes

Time	Topic	Content and Activities	Resources
3 mins	Introduction		
5 mins	What are hypotheses?	- It's a testable statement proposing a relationship between things. - It needs to be testable – there has to be a pathway from hypothesis to truth. - It is something that is provisional – you shouldn't state it as knowledge until it has been tested. - They are used to drive analysis.	

Time	Topic	Content and Activities	Resources
		Example: 'Actor X is targeting the energy sector' - testable (look for targeting activity), provisional (might be wrong), drives analysis (tells us what to collect and analyse).	
5 mins	Generating hypotheses	- A hypothesis needs to come from somewhere. - You need to know where your hypothesis has come from. - General knowledge hypotheses → Good for starting collection, broad exploration - Inductive hypotheses → Evidence-based but need testing against new cases - Deductive hypotheses → Theory-driven, generate specific predictions - Abductive hypotheses → Explain anomalies, creative thinking	
5 mins	Testing hypotheses	- Identify what evidence would support the hypothesis. - Identify what evidence would refute the hypothesis. - Build information needs (what questions to answer). - Collect and analyse. - Revise, refine, or reject.	
10 mins	Exercise: Build information needs	- Students are given a narrative about a situation. - They are given a hypothesis of what might be going on. - From that, they are to generate information needs (3-5). - As a cohort, we build these into a small intelligence requirement. - Instructor emphasises: good information needs are specific, answerable, and tell you how to test the hypothesis.	
2 mins	Closing	Tell 'em what you told 'em.	

Session Plan 9: Structured Analytic Techniques

Course Title	Intelligence Tradecraft		
Session Title	Structured Analytic Techniques		
Purpose	This session provides practical tools for overcoming cognitive biases and managing analytical uncertainty. Students will be able to identify hidden assumptions in analysis, systematically test competing hypotheses using evidence, and explore alternative future scenarios.		
Learning Outcomes	Apply basic structured analytic techniques		
Resources & Equipment	PowerPoint (live presentation) Moodle (exercises) A Tradecraft Primer: Structured Analytic Techniques for Improving Intelligence Analysis: US Government Document		
Preparation	Moodle exercise check; class admin. Upload reference library to Moodle.		
Participants	12	Session Duration	60 minutes

Time	Topic	Content and Activities	Resources
3 mins	Introduction	- What we will be covering. - How it links to earlier sessions.	
10 mins	What are structured analytic techniques?	- Structured analytic techniques are tools to help analyst to overcome challenges such as complexity, incomplete information, and the ways that we think. They are researched, documented, and applied tools that analysts can deploy against a range of domains. - Four categories overview (Diagnostic, Contrarian, Imaginative, plus implementation).	A Tradecraft Primer: Structured Analytic Techniques for Improving Intelligence Analysis

Time	Topic	Content and Activities	Resources
		Go through the range of techniques, with short descriptions of each.	
15 mins	Key assumption check	- Introduce the technique Key Assumption Check. - Go back over our inductive and deductive reasoning sections, the importance of knowing the level of certainty of your information and conclusions. - Reexamine cognitive biases and heuristics. - Learn how to identify and extract assumptions. - Run through the formal method. Exercise: extract key assumptions from analytic product. Students will be given analytic product by the instructor and will need to identify the assumptions in it.	
10 mins	Analysis of competing hypotheses	- Introduce the technique ACH. - Go briefly over hypothesis generation (was previous module). - Demonstrate how to generate competing hypotheses, including a negative hypothesis. - Demonstrate how to use the simple matrix. - Show more complex versions, including those which consider admiralty codes. - Give examples of when this can be used (and where I have used it) across different intelligence missions.	
10 mins	Exercise: Analysis of Competing Hypotheses	- Give students a series of intelligence products (5 items). - Give them hypotheses for the exercise (3 hypotheses, including 1 negative). - Have them fill in their ACH matrix. - Return to class to assess the likely hypothesis. - Remind them that this is for testing, not for a definitive conclusion.	
10 mins	What If? Analysis	- Introduce the technique What If? Analysis. - Talk about the challenges of future assessments. - Run through the method. Exercise: Have students break into small groups to examine a scenario. - As a class, go through the scenario. - This is meant to be lighter and more creative, let it run like that.	
2 mins	Closing	- Go over resources available again. - Tell 'em what you told 'em.	

Session Plan 10: Information Reports

Course Title	Intelligence Tradecraft		
Session Title	Information Reports		
Purpose	This session teaches students how to transform raw collection into structured, fact-based information reports using the WWWHW framework, creating foundational intelligence products that separate observation from analysis.		
Learning Outcomes	Write basic information reports. Understand how to use who what where when how why & whither in analysis.		
Resources & Equipment	PowerPoint (live presentation) Moodle (exercises) Example collect to use for information reports.		
Preparation	Moodle exercise check; class admin.		
Participants	12	Session Duration	30 minutes

Time	Topic	Content and Activities	Resources
3 mins	Introduction	Introduce the module.	
5 mins	What are information reports?	- An information report is a structured document that presents collected information in an organized, accessible format without necessarily drawing analytical conclusions or making assessments. - Descriptive rather than analytical - "what we know" not "what it means". - Fact-based - presents information extracted from collection. - Structured - organized around a framework (like WWWHW).	

Time	Topic	Content and Activities	Resources
		- Sourced - indicates where information came from and its reliability. - Foundational - provides the raw material for subsequent analysis. - They are an intelligence product but are different from intelligence reports.	
10 mins	How to write information reports	- Instructor gives a structure. - Begin with time. - Use subject-action of interest/event-object or at location. - Start with most important or first in sequence. - Order them in BLUF format. - Keep content true to the source. - Move through subsequent events, relationships, locations etc. using WWWHHW. - Then go to stated motive and method. - Do not speculate, derive information only from the source content. - The Admiralty Code will match the source assessment.	
10 mins	Exercise: Write information reports	- Students are given raw intercepts. - These will be simple, things like tweets and non-editorial reporting. - They are to create three information reports following the format.	
2 mins	Closing	Tell 'em what you told 'em.	

Session Plan 11: Communicating Uncertainty

Course Title	Analytic Tradecraft		
Session Title	Communicating Uncertainty		
Purpose	Students will be able to assess their own analytical confidence, communicate uncertainty appropriately to stakeholders, and use words of estimative probability to convey qualified judgments that support decision-making.		
Learning Outcomes	Understand and use words of estimative probability. Communicate uncertainty to decision makers		
Resources & Equipment	PowerPoint (live presentation) Moodle (exercises) Canned intelligence examples for exercises, uploaded to Moodle.		
Preparation	Moodle exercise check; class admin.		
Participants	12	Session Duration	50 minutes

Time	Topic	Content and Activities	Resources
3 mins	Introduction	- Welcome back from lunch. - What we're going to cover.	
10 mins	From information quality to analytic confidence	- Revise the information quality lesson. - Reinforce the lessons on inductive and deductive reasoning. - Consider other factors: - Is my collection comprehensive? - Have I asked the right questions?	

Time	Topic	Content and Activities	Resources
		• Are there alternative explanations? • What assumptions am I making? • Could I be wrong? • Your confidence will be based on the quality of the information and the analytic process, plus the other factors. • A structured analytic technique can increase analytic confidence. • Remember: It is difficult to know anything with certainty, especially in intelligence. What we do is understand our level of confidence and express it so that decision makers can reduce their level of uncertainty.	
15 mins	Exercise: Assess Analytic Confidence	• Students are given an intelligence report. • Students are given the sources for the report. • Students are given the analytic process the analyst went through. • They are given a proforma to assess the intelligence for: reasoning methods; source credibility; information quality; gaps; missing hypotheses; other factors. • They are then to make a judgement on how they'd express that intelligence to a decision maker. • There will be things that will be consistent, but then the assessment will be plucked out of the air, they need to pick this up.	
10 mins	Communicating uncertainty to stakeholders	• Your stakeholders are not intelligence professionals. • You will be under pressure to deliver advice because they are under pressure. • You will get told things like "what use is intelligence, you don't know anything". • Using clear and consistent language (estimative probability). • Communicating the methods as well as the results. • BLUF and from executive summary to details – you can put the details of how you came to your conclusion further down. • Building trust in intelligence and demonstrating its utility is one of the roles of the analyst and intelligence manager.	
10 mins	Exercise: Writing uncertainty	• Students are given the results of analysis. • Students are given sources of the intelligence. • They are to write briefing points (dot point) with a word of estimative probability that is appropriate.	

Time	Topic	Content and Activities	Resources
		• They are to make likely, possible, and not possible assessments, three about the same topic.	
2 mins	Closing	• Tell 'em what you told 'em. • Prepare them for capstone exercise.	

Session Plan 12: Capstone Exercise

Course Title	Intelligence Tradecraft		
Session Title	Capstone exercise		
Purpose	Students will synthesise Day 1 and Day 2 learning by executing the full intelligence cycle: collecting and extracting information, building ontological structure, applying analytical reasoning, and communicating findings in professional intelligence reports with appropriate confidence levels.		
Learning Outcomes	Explain the Data-Information-Knowledge-Wisdom framework and apply it to intelligence problems. Understand the fundamentals of ontology. Build a basic intelligence ontology and use it to capture target knowledge.		
Resources & Equipment	PowerPoint (live presentation) Moodle (exercises) Target cards and descriptions Scenario: Criminal intelligence		
Preparation	Moodle exercise check; class admin.		
Participants	12	Session Duration	120 minutes

Time	Topic	Content and Activities	Resources
3 mins	Introduction	What we're doing in this session.	
10 mins	Revision of previous topics	We're going to go over some of the theory from yesterday.	

Time	Topic	Content and Activities	Resources
		• DIKW. • WWWWHW. • Ontology. • Admiralty codes and words of estimative probability. • Induction and Deduction (just finished so simply acknowledge).	
5 mins	Introduction to the exercise	• We're going to take some raw intelligence and step it up from data and information to become target knowledge. • Instructor demonstrates turning data to information, extracts ontology, builds knowledge, and creates a target card. • Gives instructions and sets the program off.	
20 mins	Exercise: Data to information	• Students are broken into groups of 3. • Students are given collection: images, transcripts, written reports, target cards. • They are to have the sources listed. • They are to use WWWWHW to extract information from this collected information.	
20 mins	Exercise: Write information reports	• Students are to take the information from raw intelligence and produce information reports. • They should select pieces that are critical information for the later report. • In each group, each student will need to write 2-3.	
5 mins	BREAK		
10 mins	Exercise: Building an ontology	• Students are to take the information points that have been generated. • Students are to propose objects like people, locations, events, times, and build categories for fields in an ontology for this exercise. • Students are to propose relationships. • Students are then shown the actual ontology for this	
10 mins	Exercise: Information to knowledge	• Students are to take their captured information and form conclusions about the target. • They will be given some wisdom, some target specific knowledge about how things relate to each other (relations to locations, things like "a boss will never visit a targeted site before it happens" kind of thing). • These are to use inductive and deductive reasoning. • They are to explicitly state what their reasoning patterns are. • They are to state the level of certainty of their conclusion based on the source and the reasoning.	

Time	Topic	Content and Activities	Resources
10 mins	Exercise: Capturing target knowledge	- Students are to be given blank target and relationship cards. - They are to take the target knowledge they generated and record them in cards. - This will be reasonably fast, then we bring the group back together.	
20 mins	Exercise: Write short report	- Students are given a report structure. - They are to write a short report on the target: introduction, key points, and assessment. - They are to use words of estimative probability to express confidence.	
7 mins	Closing	Close off the course!	